

Gradient descent does not preserve a trainable PING loop

exp049 · 9 June 2026

The trained networks this entry uses are produced once in the shared training hub, exp022 (Training), and reused here rather than retrained.

Abstract

exp025 freezes the recurrent conductances W^{EI}, W^{IE} at biophysical values. When they are trainable, Adam does *not* preserve or recover effective PING from any tested initialisation (canonical, zero, or 10% of canonical): E→I recruitment weakens or remains absent, inhibitory firing collapses, and the network moves toward dense E firing at $\approx 90\%$ accuracy, close to the frozen PING control ($\approx 91\%$). The matrices store non-negative conductance magnitudes; inhibition is produced by the GABA reversal potential, not by a negative W^{IE} . PING is therefore a structural prior imposed by the frozen loop in this setup, not one gradient descent recovers on its own.

Method

Architecture. $N_E = 1024$ excitatory, $N_I = 256$ inhibitory, mem-mean readout, Dale’s law enforced. Hyperparameters match the exp025 PING baseline: Adam at $\text{lr } 4 \times 10^{-4}$, batch 256, surrogate slope 1, $W_{\text{in}} \sim \mathcal{N}(1.2, 0.12)$ at 95% sparsity, gradient norm clipped to 1.0, $\Delta t = 0.1$ ms, $T = 200$ ms, no firing-rate regulariser.

Both recurrent matrices are non-negative conductance magnitudes. After each optimiser step they are projected onto the non-negative cone. Their physiological sign is supplied by the pathway reversal potential: I→E contributes $g_I(E_I - V)$ with $E_I = -80$ mV and is therefore inhibitory despite $W^{IE} \geq 0$.

Sweep. Four conditions $\times$ three seeds (42, 43, 44) on the 10% MNIST subset (5600 train / 1400 test), 50 epochs. Only the initial (W^{EI}, W^{IE}) and the trainable-or-not flag vary:

Condition	W^{EI}, W^{IE} init	Trainable?
<i>frozen_ping</i> (control)	canonical biophysical	no
<i>trainable_ping_init</i>	canonical biophysical	yes
<i>trainable_zero_init</i>	0 (COBA-equivalent start)	yes
<i>trainable_small_init</i>	$0.1 \times$ canonical	yes

Canonical biophysical means $W^{EI} \sim \mathcal{N}(1.0, 0.1)$ μS , $W^{IE} \sim \mathcal{N}(2.0, 0.2)$ μS at $N_I = 256$, fan-in-normalised, so the trainer reports per-edge means of ≈ 0.0010 and ≈ 0.0078 . “PING is on” means the inhibitory loop is active: E recruits I, and I paces a gamma rhythm through GABA conductance. “The loop is lost” means E→I recruitment is too weak to sustain that regime, I activity is low or absent, and E fires densely. The firing rates and rhythmicity are the cleanest read of which regime a trained network reaches.

Results

The whole sweep collapses to one picture. In the (E firing rate, I firing rate) plane, a network “found PING” if it sits in the low-E / high-I corner, and “lost the loop” if it sits in the dense-E / silent-I corner.

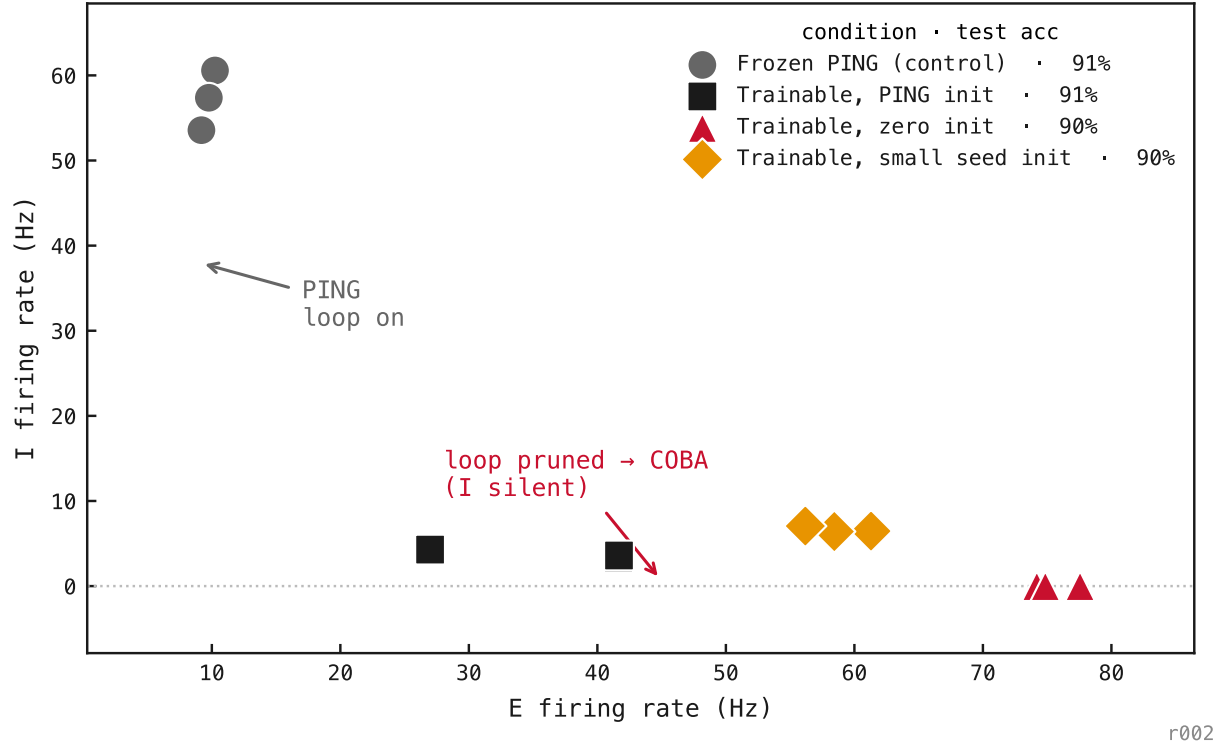


Figure 1: Each point is one trained network (4 conditions $\times$ 3 seeds). The **frozen control** (grey) sits in the PING corner, $E \approx 10$ Hz and $I \approx 57$ Hz, the inhibitory loop alive. **Every trainable condition** (started at canonical PING, at zero, or at $0.1 \times$ canonical) collapses to the dense-E / silent-I floor, $E \approx 40$ – 75 Hz and $I \approx 0$ – 7 Hz, at $\approx 90\%$ accuracy, essentially matching the $\approx 91\%$ control. Where the loop *starts* makes no difference; only whether it is allowed to train. Gradient descent never reaches PING, and dropping it costs no accuracy.

Training curves — the loop pruned, epoch by epoch

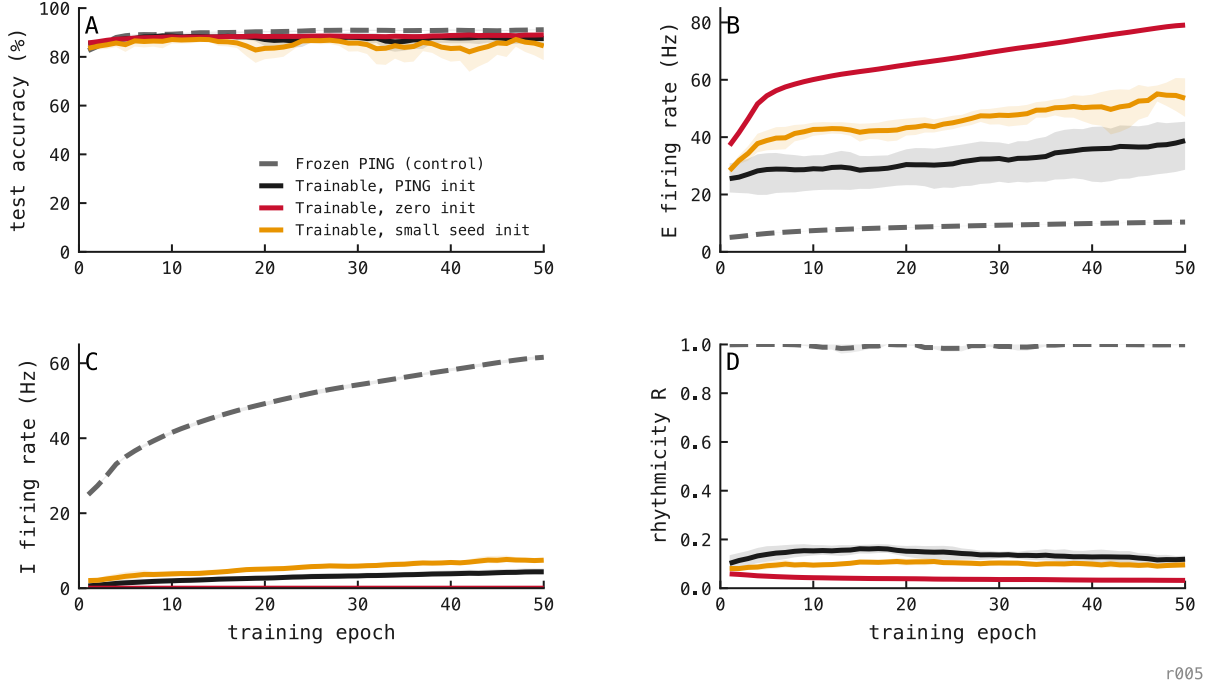


Figure 2: Per-epoch metrics from the runs themselves, coloured by init: PING init (black), zero init (red), small-seed init (amber), frozen-PING control (grey dashed); 10% of MNIST, 50 epochs, 3 seeds each. **Accuracy:** all reach $\approx 85\text{--}90\%$ and track each other; pruning the loop costs nothing. **E rate:** the trainable runs climb to $\approx 37\text{--}75$ Hz as the loop releases the excitatory population, while the frozen control stays gated near 10 Hz. **I rate:** every trainable run drops to ≈ 0 within a few epochs (the inhibitory loop is pruned), while the frozen control's I rate *rises* to ≈ 57 Hz as its readout trains. **Pingness:** the exp054 lobe–trough contrast holds near ≈ 0.99 for the frozen control while every trainable init collapses to $\approx 0.1\text{--}0.2$. The residual floor is the metric's known low-rate inflation under shared input (exp054), not a surviving rhythm: the frozen-vs-trainable gap is the signal. The loop dies early, from every start, with no accuracy penalty.

Phase portrait — there is only one attractor under training

The four per-epoch panels above tell the story but separate the two state variables that matter. Putting *E rate* and *pingness* on perpendicular axes shows the trajectories of training directly, and the geometry rules out a misreading: it is *not* the case that PING and COBA are two attractors of the same dynamics, with the architecture deciding the basin. Under training there is **one** attractor (the COBA corner), and the PING state only persists because freezing the loop zeroes its gradients.

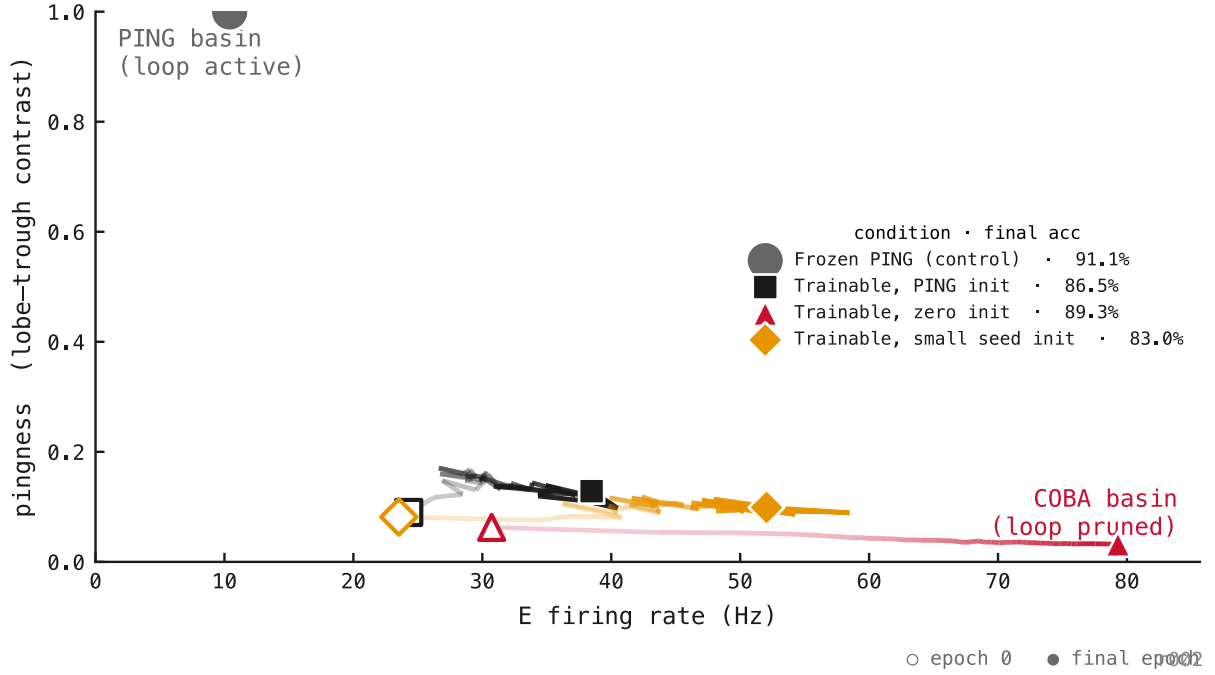


Figure 3: Each trainable trajectory is the per-epoch mean across 3 seeds, alpha-ramped along the epoch axis so the direction of flow reads off the line itself; open markers are epoch 1 (the first logged epoch), filled markers are epoch 50. **Frozen PING (grey)**: sits in the upper PING basin from start to finish at pingness ≈ 0.98 ; drifts a little in E rate ($\approx 4 \rightarrow 10$ Hz) as the feedforward and readout weights train, but the loop weights themselves can’t move by construction. **Trainable PING init (black)**: was initialised with canonical biophysical loop weights, but by the time the first metric is logged (after epoch 1) the loop has already been pruned to pingness ≈ 0.17 . The trajectory then walks rightward along the COBA floor with the others. **Trainable zero init (red)** and **small-seed init (amber)**: start at the floor and stay there, E rate climbing as the feedforward layer learns the task. Every legend entry lands at 83–91% accuracy, so the move costs nothing. The reading is sharper than the time-series panels suggest: the empty band between pingness ≈ 0.2 and ≈ 0.85 is the *whole story*, because gradient descent flies through it within one epoch and the metric never catches it mid-flight. There is no separatrix between PING and COBA under training because there is no slow trajectory through the gap. The freeze isn’t preventing slow erosion; it’s preventing instant collapse.

Accuracy–rate trajectory — same destination, different spike economy

Figure 3 puts pingness on its own axis. Putting pingness on the *colour* axis instead and replaying training as (E rate, accuracy) trajectories (the exp025 accuracy–rate frontier given a time axis) gives the third reading of the same data: every condition reaches the same final test accuracy, but along very different routes, and only one of them is doing PING while it gets there.

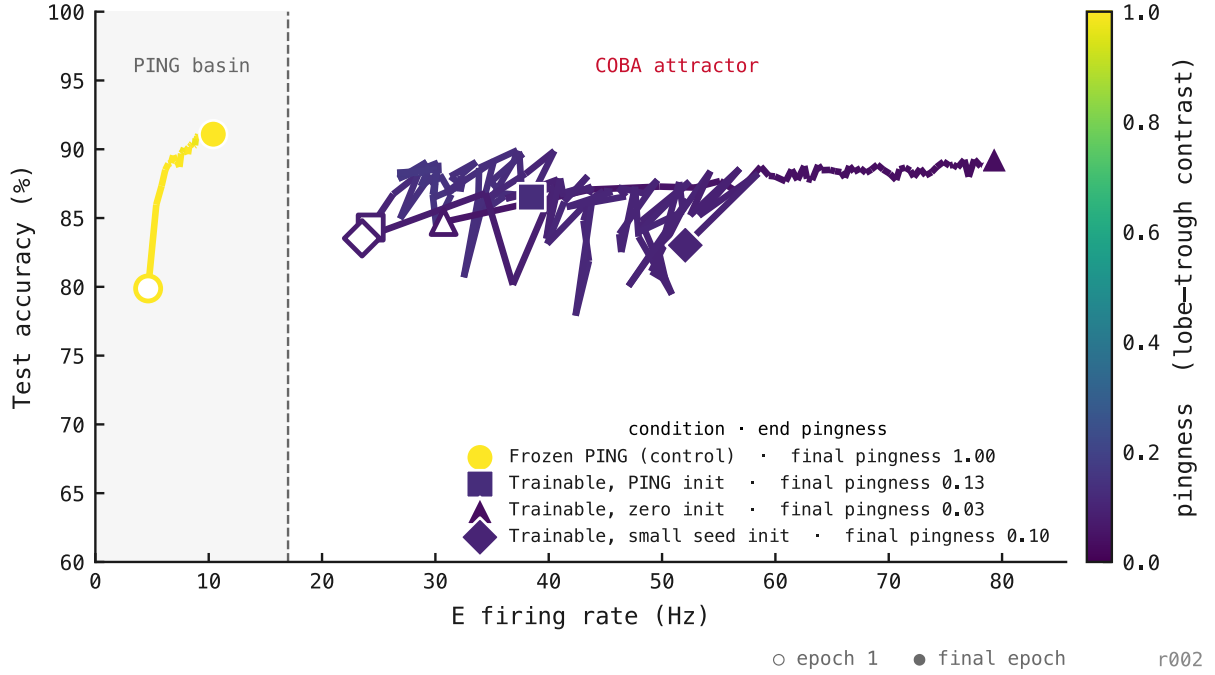


Figure 4: Each trajectory is the per-epoch mean across 3 seeds; per-segment colour is that epoch’s pingness on a viridis 0→1 scale (colourbar at right). Open markers are epoch 1, filled markers are epoch 50. The frontier *with time* tells three things at once. **Same destination:** all four conditions land at $\approx 83\text{--}91\%$ accuracy by epoch 50, regardless of init or whether the loop trains. **Different routes:** the frozen control climbs accuracy almost vertically with little change in E rate ($\approx 4 \rightarrow 10$ Hz); the trainable conditions all bend right and up, the biggest one (zero init) reaching final accuracy at $E \approx 75$ Hz, $\approx 8\times$ the frozen control’s rate. **Only one of them is still PING:** the frozen control’s line is bright yellow because pingness stays high (≈ 1.0) the whole way; every trainable line is dark purple because pingness was already pruned to ≈ 0.1 by the first logged epoch. Reading the colour bar: most of pingness 0.2–0.85 is unused space, because no trajectory crosses it slowly enough to be logged. Reading the geometry: this is the manuscript’s accuracy–rate frontier (exp025, §2.3) animated, and the architecture is the only thing keeping a trained network on the sparse, rhythmic side of it. The dashed divider at ≈ 17 Hz marks the two basins explicitly: frozen PING holds the left, every trainable run ends in the right, at the same accuracy.